Alice Gao

PhD Researcher

Human-computer interaction • UI Understanding • Computational Interfaces • Human-AI interaction • Data-driven design tools

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Research Experience

Research Assistant / University of Washington

Jan 2024 - ***Present*** · Seattle, WA

- Developed novel machine-learning algorithm to train computer use agents (CUAs) to automatically perform automated usability testing, increasing CUA usability assessment accuracy by up to 45%.
- Synthesized automatic UI design metrics across literature to develop a new method of understanding UI design.
- Developed automatic pipeline to automatically compute design metrics for 10,000 collected websites across the globe and integrated into a research prototype.
- Led 10 user studies with UI/UX designers using a research prototype demonstrating new method of understanding UI design provides confidence and enables designers to provide more objective descriptions of design.
- Designed new interventions to reduce AI overreliance during AI-supported writing by up to 30% with 200+ participants in an online experimental study.

GRAIL / University of Washington

Sep 2021 - Jan 2024 · Seattle, WA

- Developed machine learning models to redirect attention and eye gaze for a more attentive video conferencing experience

Industry Experience

Capital One / Software Engineering Intern

Jun - Aug 2020 · New York City, NY

Created new endpoint and unit tests, laid framework for streaming data platform, and launched 2 new security groups.

Tezign / AI Design Lab Intern

Jun - Aug 2019 · Shanghai, China

Assisted in optimizing image retrieval to match similar designs and contributed to training a model in judging strong vs weak graphic designs for advertisements.

Education

University of Washington

Sep 2021 - *Expected Dec 2027*

MSc in Computer Science *(2021 - 2023)*
PhD in Computer Science (2023- **Present**)

Princeton University

2017 – 2021

B.S in Computer Science

Selected Publications

Alice Gao, Weixi Tong, Rishab Vempati, Katharina Reinecke, R. Benjamin Shapiro, Tianyi Zhang, Jason Wu. Training Computer Use Agents to Assess the Usability of Graphical User Interfaces. April 2026. arxiv.org/pdf/2604.26020.

Alice Gao, Andrew N. Meltzoff, Maarten Sap, Katharina Reinecke. Framing an AI with Values Reduces AI Reliance in AI-supported Writing Tasks. ACM Conference on Fairness, Accountability, and Transparency (FAccT '26). Montreal, QC, Canada. June 2026. doi.org/10.1145/3805689.3812369.

Skills & Languages

Skills:
Machine learning · Post-training · Interviews (in-person and remote) · Survey Design · Usability testing · Thematic analysis · Statistical modeling · Data Analysis · Experimental design

Languages:
Python · R · Javascript · SQL · TypeScript

Frameworks / Tools:
PyTorch · scikit-learn · Pandas · React · Sveltekit · Flask · FastAPI · Figma